Atif ANWER

Computer Vision Researcher,
AI & Robotics Engineer

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Atif-Anwer
atifanwer

Education

2019 – 2022

Ph.D. Computer Sciences / Electrical & Electronics Engineering *(Co-tutelle)*,
INSA Rouen Normandie, France & University Teknologi PETRONAS, Malaysia,
Funded by: Agence Nationale de la Recherche (ANR) project ICUB,
Supervisors: Prof. Samia Ainouz (INSA), Asst. Prof. Naufal Bin Saad (UTP)
Thesis: Specular Highlight Detection and Adversarial Generation of Specularity-Free Images using Multi-Domain Translation Network Trained with Polarimetric Data

2016 – 2017

M.Sc. Electrical & Electronics Engineering,
University Teknologi PETRONAS, Perak Malaysia,
Supervisor: Asst. Prof. Syed Saad Azhar Ali
Thesis: Real-time underwater 3D scene reconstruction using Kinect v2 time of flight camera

2001 – 2004

B.E. Mechatronics,
Major: Robotics, Control Systems and Image processing,
National University of Sciences & Technology (NUST), Pakistan,
Supervisor: Dr. Saeed Ur Rahman
Thesis: Development and control of Bipedal Autonomous Robot (BART)

Interests

- Developing problem specific Object detection and semantic segmentation models
- Modeling, design, and end-to-end development of mobile robotic systems integrating DFAM CAD modeling, perception, control, and embedded intelligence
- Human pose estimation for motion analysis and real-world deployment
- Generative modeling (GANs, VAEs, diffusion models) for data augmentation and synthetic data generation

Technical & Professional Highlights

- Advanced proficiency in Python for image processing pipelines, deep learning model integration and deployment workflows.
- Strong background in multimodal data processing including imaging datasets for medical image analysis, segmentation and biomarkers detection within collaborative research environments.
- In-depth knowledge and understanding of Mechatronic systems, 3D modeling, 3D printing, hardware integrations and robotic systems.
- Experienced in scientific communication, preparation of technical documentation, research reports, oral presentations, and interdisciplinary collaboration with clinicians, engineers, and academic peers.

Experience

2024 **Postdoctoral Researcher**, *Université Bourgogne Europe*, Dijon, France,

Research and development of software pipeline and Web Application for end-to-end detection, segmentation, and analysis of Tunnelling Electron Microscopy (TEM) images of Cu, Ni and ZnO nanoparticles. Additionally, development on estimating risk of cardiovascular disease from en-face OCT fundus images using deep learning.

2022–2024 **Machine Learning R&D Engineer**, *Apeira Technologies*, Le Creusot, France,

Research and development of deep learning algorithms for human pose estimation and posture correction, to improve cyclist performance in videos.

2018–2020 **Freelance 3D CAD & Graphic Designer**, *Freelancer.com*

Executed multiple 3D CAD modelling and graphic design projects from concept to prototype to completion.

2015–2016 **Co-Founder & CTO**, *Scifacterz*, Islamabad, Pakistan

Robotics Startup venture centred on design and of "NBE (Non-Biological Entity)" an open-source, low cost, research platform and Home Service Robot

2004–2016 **Manager (Mechatronics)**, Islamabad, Pakistan

Multiple diverse industry roles involving concurrent responsibilities in Mechanical and Electronic Design research, and managerial positions.

Skills

Languages & Frameworks	Python, C, MATLAB, TensorFlow, PyTorch, OpenCV, React, Electron, Typescript
Computer Vision & Robotics	Object detection, Pose estimation, 3D vision & perception, Sensor integration, Mobile Robot design and end-to-end prototyping, development, ROS2
LLMs & Generative AI	Multimodal AI (vision-language models, CV + LLM pipelines), Fine-tuning (LoRA etc.), Prompt Engineering, Agentic Coding
Data & ML Engineering	Dataset curation, Gradio for ML prototyping and deployment
3D Modeling & Design	Creo, SolidWorks, Blender, SketchUp, Adobe Photoshop, Affinity Designer, Figma
Tools & Platforms	Git, LaTeX, Notion, WordPress; 3D printing and DFAM workflows

Languages

Urdu	Native	
English	Fluent	*IELTS (Band 8.0)*
French	Medium	*Good Understanding, Medium Speaking*

Extra Curricular/Pro-Bono

2020 **Administrator**, r\ComputerVision subreddit and Computer Vision discord server

2010–2011 **Guest writer (Hardware reviewer)**, *wccftech.com*

2007–2012 **Gaming Server Lead Administrator**, *Micronet Broadband Ltd.*, Islamabad, Pakistan

Publications

Anwer, Atif, Paul Fleurat-Lessard, Zhiling Qian, Fabrice Mériaudeau, Jean-Cyrille Hierso, and Myrtil L. Kahn. *TAUTEM-2D - An end-to-end solution for nanoparticle analysis from TEM images*. In *IEEE International Conference on Image Processing (ICIP) - (Submitted)*, Feb 2026.

Lucas Szwarcberg, **Anwer, Atif**, Alexandre Gozlan, Paul Drouard, Pierre-Henry Gabrielle, Catherine Creuzot-Garcher, Fabrice Mériaudeau, and Louis Arnould. The aroma dataset for automatic detection of artifact type and severity in retinal optical coherence tomography angiography. *Ophthalmic Research (Accepted)*, 2026.

Louis Arnould, **Anwer, Atif**, Andrzej Grzybowski, and Fabrice Mériaudeau. Selecting the right AI algorithm for the job: A guide for navigating the AI jungle in ophthalmology. *Ophthalmol. Ther.*, 14(8):1637–1647, August 2025.

Pétra Eid, Abderrahmane Bourredjem, **Anwer, Atif**, Catherine Creuzot-Garcher, Pearse Andrew Keane, Yukun Zhou, Siegfried Wagner, Fabrice Mériaudeau, and Louis Arnould. Retinal microvascular biomarker assessment with automated algorithm and semiautomated software in the montrachet dataset. *Translational Vision Science and; Technology*, 14(3):13, March 2025.

C. Germanese, **Anwer, Atif**., P. Eid, L.-A. Steinberg, C. Guenancia, P.-H. Gabrielle, C. Creuzot-Garcher, F. Mériaudeau, and L. Arnould. Artificial intelligence-based prediction of neurocardiovascular risk score from retinal swept-source optical coherence tomography–angiography. ***Scientific Reports*** *(IF: 3.8)*, 14, Nov 2024.

Anwer, Atif., S. Ainouz, N M Saad, S S A. Ali, and F. Mériaudeau. *Joint network for specular highlight detection and adversarial generation of specular-free images trained with polarimetric data*. ***Neurocomputing*** *(IF: 5.5)*, 559:126769, 2023.

C. Germanèse, F. Mériaudeau, P. Eid, R. Tadayoni, D. Ginhac, **Anwer, Atif**., S. Laure-Anne, C. Guenancia, C. Creuzot-Garcher, P. Gabrielle, et al. *A Retinal Oct-Angiography and Cardiovascular STAtus (RASTA) Dataset on Swept-Source Microvascular Imaging for Cardiovascular Risk Assessment*. ***Data*** *(IF: 2.2)*, 2023.

Anwer, Atif, Samia Ainouz, Mohamad Naufal Mohamad Saad, Syed Saad Azhar Ali, and Fabrice Mériaudeau. *SpecSeg network for specular highlight detection and segmentation in real-world images*. ***Sensors*** *(IF: 3.4)*, 22(17):6552, 2022.

Anwer, Atif, Syed Saad Azhar Ali, Amjad Khan, and Fabrice Mériaudeau. *Underwater 3D scanning using Kinect v2 time of flight camera*. In *Thirteenth International Conference on Quality Control by Artificial Vision 2017*, volume 10338. SPIE, 2017.

Anwer, Atif, Syed Saad Azhar Ali, Amjad Khan, and Fabrice Mériaudeau. *Underwater 3D Scene Reconstruction Using Kinect v2 Based on Physical Models for Refraction and Time of Flight Correction*. ***IEEE Access*** *(IF: 3.4)*, 5:15960–15970, 2017.

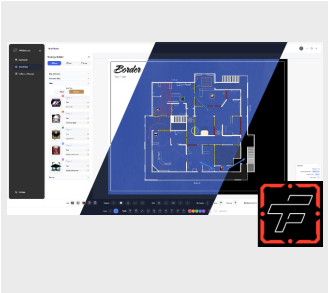
Anwer, Atif, Syed Saad Azhar Ali, Amjad Khan, and Fabrice Mériaudeau. *Underwater 3D scanning using Kinect v2 time of flight camera*. In *The International Conference on Quality Control by Artificial Vision 2017*, 2017.

Anwer, Atif, Fabrice Mériaudeau, and Syed Hasan Adil. *Customized graphical user interface implementation of Kinect Fusion for underwater application*. In *2017 IEEE 7th International Conference on Underwater System Technology: Theory and Applications (USYS)*, pages 1–6. IEEE, 2017.

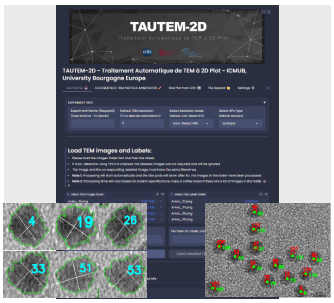
IF: Clarivate Web Of Science Impact Factor

Select Project Portfolio

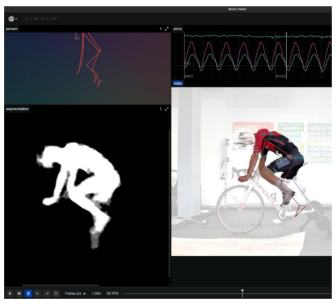
Development of a ROS-2 based holonomic autonomous mobile robot with embedded motor control (STM32, Raspberry Pi 5), LiDAR for mapping, and network-offloaded modular vision perception (YOLO and Mediapip). (2026) (In-Progress)



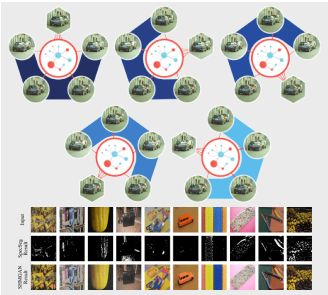
Sole developer and maintainer of FPSTACTICIAN, an interactive Strategy and planning Windows application for the Ubisoft game Rainbow Six SiegeX. Tech stack includes Electron (platform), React (frontend), Type Script and TailwindCSS with Supabase for Authentication. (2025) (FPSTactician.com)



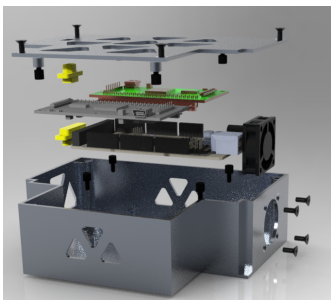
Development of robust end-to-end detection, segmentation, and analysis pipeline of Tunnelling Electron Microscopy (TEM) images of nanoparticles, using a custom-trained YOLO v8. Implemented as a interactive Gradio user interface for seamless navigation and control of the project. (2026) (ICIP-Submitted)



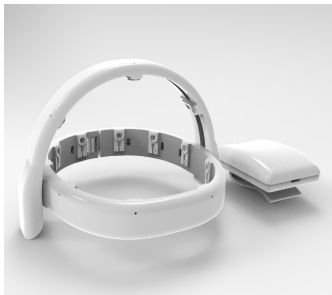
Development of real-time human pose estimation algorithms for cyclist performance improvement and posture correction onboard Android mobile devices, utilizing MediaPipe and transfer-learning to custom GCNs to generate accurate SMPL-X meshes. (2024)



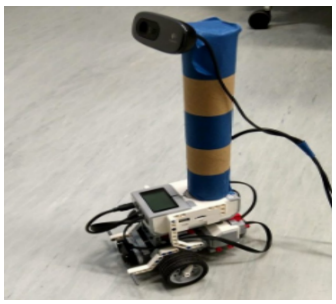
A multi-input Generative Adversarial Network (GAN) to generate specular-free images from a single RGB input image without external labels, trained using polarimetric images and showcased performance in challenging real-world scenarios (2023) [Neurocomputing] [Github]



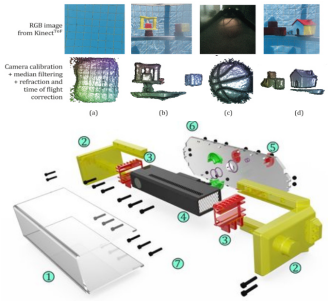
NASA Electronics Box Design Competition (2020) *[Runners Up]*. Designing a lightweight electronics box for the **NASA Astrobee Robotic Arm contest** to accommodate individual electronics boards, connectors, and a mechanical mounting bolt pattern. (2020)



Designed, prototyped and delivered a fully modular universal portable EEG headset to **Brain9D**, utilizing OpenBCI for sensing using dry electrods and a portable electronics housing for ergonomic design (2020)



QR-code–following robot using real-time detection, SIFT/KLT tracking, and PID control, implemented in MATLAB on a Lego Mindstorms platform for and obstacle avoidance with monocular vision (2017).



3D scene reconstruction using Kinect Fusion with Kinect v2 for underwater depth data, including a custom waterproof housing and C# interface with novel time-of-flight and refraction correction algorithms. (2017) [IEEE Access] [Github]

Projects

FPS Tactician (Personal Project) (2025 ➞)

Sole developer/maintainer of FPS TACTICIAN, an interactive Strategy and planning Windows application for the Ubisoft game *Rainbow Six SiegeX*, with 500+ active users. Tech stack includes Electron (platform), React (frontend), TypeScript, and TailwindCSS, with Supabase for Authentication. (FPS Tactician)

Automated analysis of TEM images of nano-particles (2023 ➞)

Solely responsible for creation of a robust software pipeline and Web Application for end-to-end detection, segmentation, and analysis of Transmission Electron Microscopy (TEM) images featuring Cu, Ni and ZnO nanoparticles, by deploying a custom-trained YOLO v8 OBB network.(TAUTEM2D Platform)

Cyclist pose estimation and posture correction (2023-2024)

Research and development of deep learning algorithms aimed at human pose estimation and posture correction, with a focus on enhancing cyclist performance in videos from hand-held devices. Solely responsible for refining MediaPipe accuracy by transfer-learning of graph-convolutional network (GCN) and utilizing the improved pose estimates to generate accurate SMPL/SMPL-X meshes, facilitating various cycling performance calculations.

Estimating risk of cardio-vascular diseases from OCT-A images (2023 - 2025)

Comprehensive research and development project on estimating the risk of cardiovascular disease from en-face OCT fundus images. Employed a Convolutional Neural Network (CNN) architecture with EfficientNet v2 as the backbone. A journal publication of the research work has been submitted for peer-review.

SpecSeg (2021-2022)

Conceived and developed a highly efficient Specular Segmentation (SpecSeg) network, structured on the U-net architecture that can detect pixels strongly affected by specular highlights with a high degree of precision.
(SpecSeg repository , Huggingface Demo)

SHMGAN (2019-2022)

PhD Research - Development of a multi-input Generative Adversarial Network (GAN) designed to produce specular-free images from a single RGB input image, eliminating the need for external labels. Specular Highlights Mitigating GAN (SHMGAN) was trained using polarimetric images obtained under uncontrolled lighting conditions and showcased adversarial network design within challenging real-world scenarios. (SHMGAN repository)

Teaching

2023–2026 **Masters Courses**, ***Image Processing (2023, 2024, 2025)***, MS Health AI (M2)), University Bourgogne Europe, France

2021–2022 **Undergraduate Courses**, ***Data Analysis and Digital Tools (2021, 2022)***, Département Informatique et technologie de l'information, INSA Rouen, France

2019 **Teaching Assistant**, *Probability & Statistics*, Department of Electrical & Electronics Engineering, UTP, Malaysia

Achievements

2019–2020 **Graduate Assistant Scholarship**,
Universiti Teknologi PETRONAS, Malaysia

2016–2017 **Graduate Assistant Scholarship**,
Universiti Teknologi PETRONAS, Malaysia

2016–2017 **1st round favourite**,
Robot Launch 2014 (Global robotics startup competition), RoboHub.org

2011 **Winner "Intel More to the Core™ Challenge"**,
Intel™, Pakistan

2010 **Winner "CPU IQ Blogger Challenge, South Asian region"**,
Intel™, Pakistan

2009 **Member Technical Evaluation & Judges panel**,
"RoboSprint 2010", Air University, Islamabad

2003 **Winner "Best Idea Award"**,
"1st National Fire Fighting Robot Competition (FFRC)", College of EME, NUST, Pakistan

Internships & Co-supervision

2025 ***Technical Co-Supervision of PhD thesis: Designing AI tools for automated information extraction and predicting optimal chemical reaction condition***,
Identifying optimal chemical reaction conditions to maximize yield by combining state-of-the-art reaction-graph neural networks with LLM-driven automation for extracting reaction knowledge from the literature and proposing next-generation alternatives to existing condition-prediction methods.,
PhD Student: Ulrick Fineddie Randriharimanamizara -
Supervisors: Julien Roger, Fabrice Meriaudeau, Paul Fleurat-Lessard,
University Bourgogne Europe, Dijon, France

2025 ***Augmented Reality in Mandibular Tumor Surgery***,
Development of a real-time AR system for mandibular tumor surgery by overlaying CT 3D reconstructions onto endoscopic video, using feature extraction, tracking and homography chaining for accurate image registration.,
University Bourgogne Europe, Dijon, France

2024 ***Fully automated detection of nanoparticles in TEM images***,
Detection of Zn, CuO and Ni nanoparticles in Transmission Electron Microscopy (TEM) images using YOLO and generation of 2D size plots based on the NP dimensions,
University Bourgogne Europe, Dijon,

References

Fabrice Mériaudeau
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ICMUB Institute of Molecular Chemistry of the University of Burgundy (ICMUB)
University de Bourgogne, France
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Laboratoire d'Informatique, du Traitement de l'Information et des Systèmes (LITIS) Lab, INSA Rouen, France
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